Week 1: miRNA-mRNA human – chimp brain development

Goals: 1.) assemble the core PFC expression and metadata resources

2.) confirm the matched subjects

1.) PFC dataset summary from article

Prefrontal cortex sample sizes and retained expression features in the human, chimpanzee and rhesus macaque comparison.

Dataset	Human n	Chimpanzee n	Macaque n	Total n	Retained PFC features
mRNA microarray	23	12	26	61	12,447 expressed genes
miRNA microarray	8	11	8	27	198 detected miRNAs

Selection notes

- mRNA: probes were restricted to sequences matching the human, chimpanzee and macaque genomes perfectly and uniquely; 12,447 genes were expressed in the PFC dataset.
- miRNA: 866 array features were reduced to 500 mature miRNAs conserved across all three species and then to 198 miRNAs detected in the PFC after expression filtering.
- The mRNA and miRNA experiments used different, only partially overlapping sets of individuals and should not be treated as a completely paired cohort.

Data downloaded (prefrontal cortex (PFC) and cerebellum (CBC))

Source	Download	Decscription
GSE29356	Processed PFC and CBC miRNA file and series metadata	Processed PFC miRNA expression. 198 detected in PFC
GPL10850	Platform annotation table	Annotations for individuals in expression data PFC and CBC
GSE22521	Processed PFC mRNA expression or series matrix	PFC mRNA expression. 12447 genes expressed in PFC
GSE22569	Processed CBC mRNA expression or series matrix	CBC mRNA expression. 12853 genes expressed in CBC
GPL6244	Platform annotation table	Map Affymetrix probes to genes
Somel et al.	Table S1 and Text S1	Sample metadata and methods

Naming note. The article cites GSE22570, the complete mRNA data. GSE22521 is its PFC component and is therefore the appropriate starting dataset for the PFC-only analysis.

Normalization note. The authors' processed expression values were log2 transformed and quantile normalized. They are suitable for learning and for reproducing downstream analyses without beginning with raw microarray preprocessing.

Data files:

- `series_matrix.txt.gz` = expression matrix;
- `family.soft.gz` = detailed metadata and documentation, sometimes including another representation of the processed values;

Sources

1. Somel M et al. MicroRNA-Driven Developmental Remodeling in the Brain Distinguishes Humans from Other Primates. PLOS Biology 9(12): e1001214. https://doi.org/10.1371/journal.pbio.1001214
2. NCBI Gene Expression Omnibus GSE29356. https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE29356
3. NCBI Gene Expression Omnibus GSE22521. https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE22521
4. NCBI Gene Expression Omnibus GPL6244. https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GPL6244
5. TargetScanHuman Release 8.0 data download. https://www.targetscan.org/cgi-bin/targetscan/data_download.vert80.cgi

Week 2: miRNA-mRNA human PFC – chimp brain development

2.) PFC mRNA sample reconciliation

Notebook used: 1a_PFC_mRNA_check_data.ipynb

Comparison of GEO GSE22521 microarray measurements, Table S1 subject annotations, and the published PFC mRNA sample counts.

Sample counts

Source / selection	Human n	Chimpanzee n	Macaque n	Total n
GEO arrays (original)	—	—	—	68 arrays
GEO after five repeat pairs are removed	23	12	28	63*
Table S1 PFC mRNA subjects	23	12	27	62
Published PFC mRNA analysis	23	12	26	61

*The 63-subject count is provisional: it assumes the five identified pairs exhaust repeated measurements.

Identified repeated measurements

Species	Age (days)	Measurement 1	Measurement 2	Batch
Human	2	GSM559193	GSM559194	b1 / b2
Human	35,770	GSM559198	GSM559199	b1 / b2
Macaque	10,220	GSM559208	GSM559209	b1 / b2
Macaque	16	GSM559215	GSM559216	b1 / b2
Macaque	2,936	GSM559223	GSM559224	b2 / b2_2

Selection note. Omit Measurement 2, as measurements are highly correlated. This convention is not a verified reconstruction of the original authors' choices.

Additional macaque sample and unresolved omission

GSM559211: rhesus macaque, superior frontal gyrus, age 1,487 days (~4.1 years). This GEO sample has no species-and-age match among the PFC mRNA subjects in Table S1; its absence from the table is unexplained. One further macaque remains unidentified: Table S1 lists 27 PFC mRNA macaques, but the publication reports 26. The excluded subject's Table S1 number and GSM accession cannot be established.

Working data and remaining checks

Original matrix: 13,042 features × 68 arrays.

Working matrix: 13,042 × 61 after removing five repeat-array partners, GSM559211 (unmatched in Table S1), and prenatal macaque GSM559214 (−15 days). This is an exploratory selection, not a reconstruction of the published 61-subject cohort.

Exploratory expression-correlation filtering

Filtering rule. For each individual, calculate the Pearson correlation of its 13,042-feature mRNA expression profile with every other individual of the same species. Exclude self-correlation, take the median, and retain samples only when their **within-species median correlation** is strictly **greater than 0.90 (90%)**.

Samples flagged by the 0.90 cutoff (*GSM559225 is prenatal, −30 days in Table S1):

Species	Age (days)	GEO sample	Median correlation
Human	2	GSM559193	0.845729
Human	19	GSM559186	0.877110
Rhesus macaque	−30*	GSM559225	0.883975

Dataset counts: 61 pre-filter samples after the explicit exclusions → **58 retained** after the 0.90 correlation cutoff (21 human, 12 chimpanzee, 25 macaque).

3.) Annotation of the filtered PFC mRNA data.

Notebook used: 1a_PFC_mRNA_check_data.ipynb

Feature-to-gene mapping using GPL6244

Selection stage	Count	Interpretation
Starting expression matrix	13,042 features	58 samples; one row per Affymetrix feature
No identified gene	354 features	N_genes = 0; exclude from primary gene matrix
Ambiguous gene mapping	772 features	N_genes >= 2; exclude from primary gene matrix
Exactly one gene per feature	11,916 features	N_genes = 1; retain for gene-level screening
Distinct identifiable genes	11,783 genes	Some genes are represented by multiple features
Genes with multiple features	**116 genes**	**Exclude all under the one-feature-per-gene rule**
Final gene-expression matrix	11,667 genes × 58 samples	One unambiguous feature per gene; no averaging

Filtering rules and quality checks

Gene mapping. The GPL6244 gene_assignment field was parsed into distinct gene symbols. Only features mapping to exactly one symbol were eligible. Genes with two or more eligible features were removed, rather than averaging their measurements. Multiple features assigned to the same gene may represent different transcripts, alternative splice variants, or measurements of different regions of the same gene. That was a conservative choice.

4.) PFC miRNA annotation

Notebook used: 2_PFC_miRNA_check_data.ipynb

Processed miRNA expression and annotation

Property	Result	Notes
GEO expression series	GSE29356	Processed prefrontal cortex (PFC) file, separate from CBC
Expression matrix	198 miRNAs × 27 profiles	Published processed subset; values are log2-scale and normalized
Species distribution	8 human; 11 chimpanzee; 8 macaque	Species coded Hsa, Ptr, Mml in processed column labels
Feature labels	Mature miRNA names	Retain original historical names, including * passenger-strand suffixes
Platform annotation	GPL10850	Platform record available for verifying historical probe-to-miRNA assignments
Missing/constant features	No exclusion based on variability	SD across profiles ranged from 0.200 to 2.547

Sample annotations and GEO accession reconciliation

The 27 processed columns encode species, brain region, and age (e.g., Hsa_PFC_3days). Comparing these names with original GEO family-SOFT metadata revealed a systematic one-day offset: processed age = original age in days + 1. Matching on species, PFC region, and this adjusted age recovered 22 profiles with one candidate GSM and five chimpanzee profiles with two GSM records (batch 1 and batch 2); none remained unmatched. Batch pairs are repeated measurements.

Processed chimpanzee profile	Original GSM pair (batch 1 / batch 2)
Ptr_PFC_1days	GSM725625 / GSM725626
Ptr_PFC_2days	GSM725632 / GSM725631
Ptr_PFC_40days	GSM725636 / GSM725635
Ptr_PFC_526days	GSM725624 / GSM725623
Ptr_PFC_16132days	GSM725678 / GSM725677

5.) Files saved during data processing

Filename	Contents / role
pfc_mrna_gene_expression.csv	Filtered mRNA gene-symbol expression matrix: 11,667 genes × 58 profiles, after excluding GSM559214 and the three correlation outliers.

pfc_mrna_gene_annotation.csv	GPL6244 feature-to-gene annotation for uniquely mapped features and traceability of retained mRNA measurements
pfc_mirna_sample_info.csv	Metadata for 27 processed PFC miRNA profiles, including age, species, and mapped GSM accessions (one or two per profile)

6.) mRNA and miRNA regression analysis in PFC and age distribution

Notebook used: 1a_PFC_mRNA_check_data.ipynb, 2_PFC_miRNA_check_data.ipynb, 3_PFC_miRNA_mRNA_regression_plots.ipynb

Model specification and interpretation

Species model. Categorical species indicators (chimpanzee reference); age model: log2(age in days + 1); combined model: both predictors, with a common age slope across species.

a.) mRNA: Exploratory linear regression of 11,667 retained PFC genes across 58 individuals (21 human, 12 chimpanzee, 25 rhesus macaque).

Explanatory power of species and developmental age

Regression model	Mean R^2	Median R^2
Species only	0.2582 (25.82%)	0.1957 (19.57%)
Age only	0.1298 (12.98%)	0.0601 (6.01%)
Species + age	**0.3824 (38.24%)**	**0.3642 (36.42%)**
Age added after species	0.1242 (12.42 pp)	0.0525 (5.25 pp)
Species added after age	0.2526 (25.26 pp)	0.1920 (19.20 pp)

Interpretation. Across genes, the mean in-sample R^2 was 25.8% for species, 13.0% for age, and 38.2% for the combined model in the 58-sample cohort. Adding age after species improved R^2 by 12.4 percentage points on average.

b) miRNA: Exploratory linear regresssion of 198 miRNAs across 27 individuals. The models predict processed log2 expression from species alone, log2(age in days + 1) alone, or both predictors (common age slope across species).

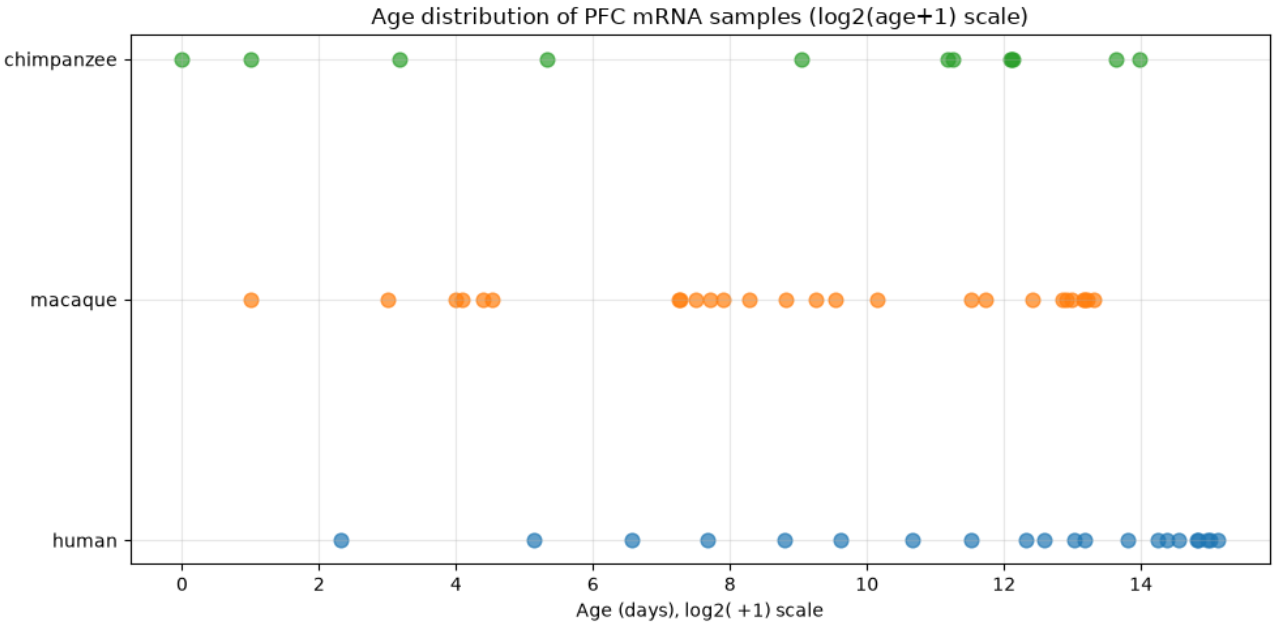
Explanatory power of species and developmental age

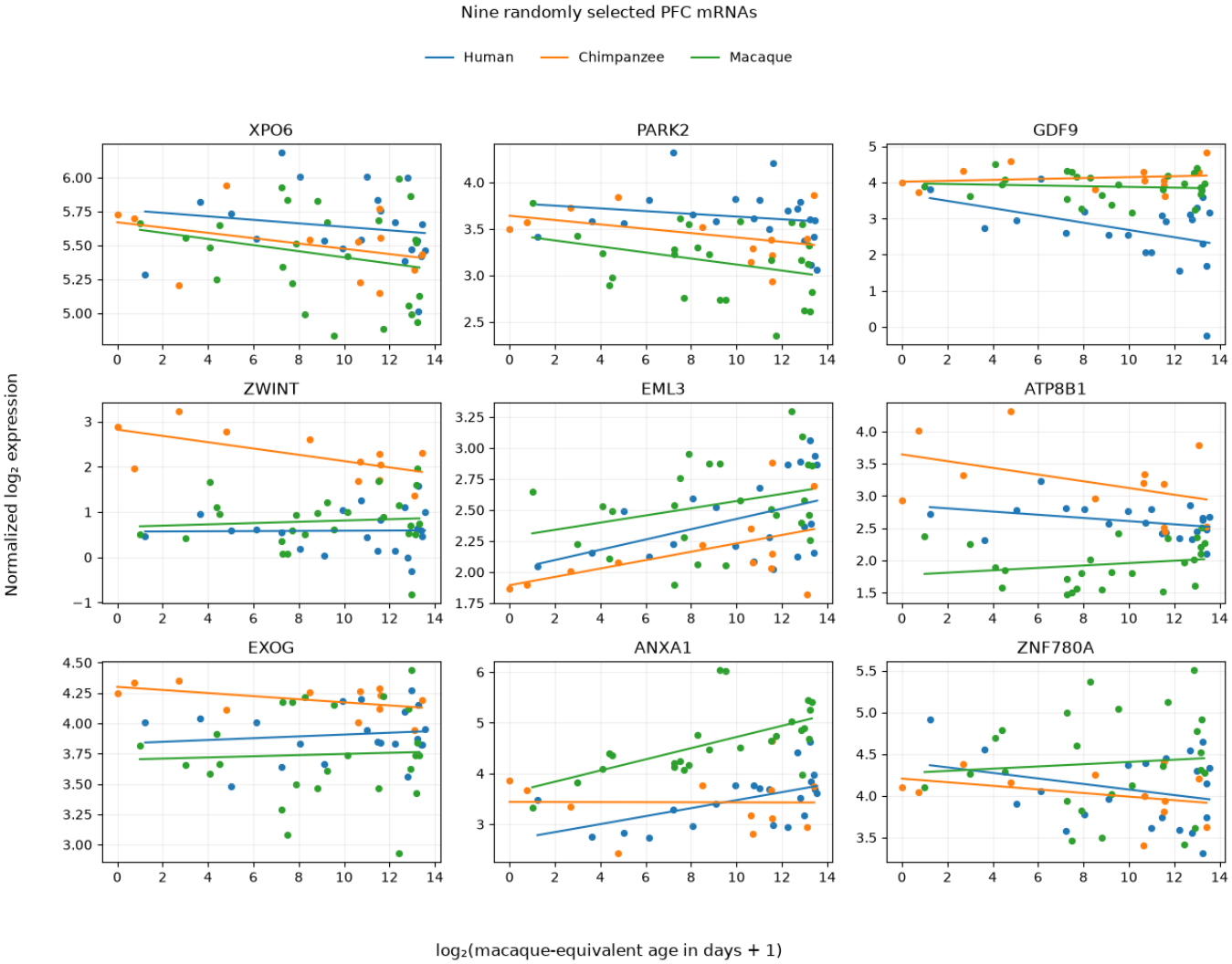
Regression model	Mean R^2	Median R^2
Species only	0.2477 (24.77%)	0.1673 (16.73%)
Age only	0.2386 (23.86%)	0.1432 (14.32%)
Species + age	**0.4966 (49.66%)**	**0.5174 (51.74%)**
Additional age after species	0.2489 (24.89 pp)	0.1652 (16.52 pp)
Additional species after age	0.2580 (25.80 pp)	0.1843 (18.43 pp)

Interpretation. Species-only explanatory power was comparable between mRNA and miRNA datasets (mean $R^2 \approx 0.258$ versus 0.248), while age was associated with more of the observed miRNA variation (0.239 versus 0.130). The combined model explained on average 49.66% per miRNA versus 38.24% per gene.

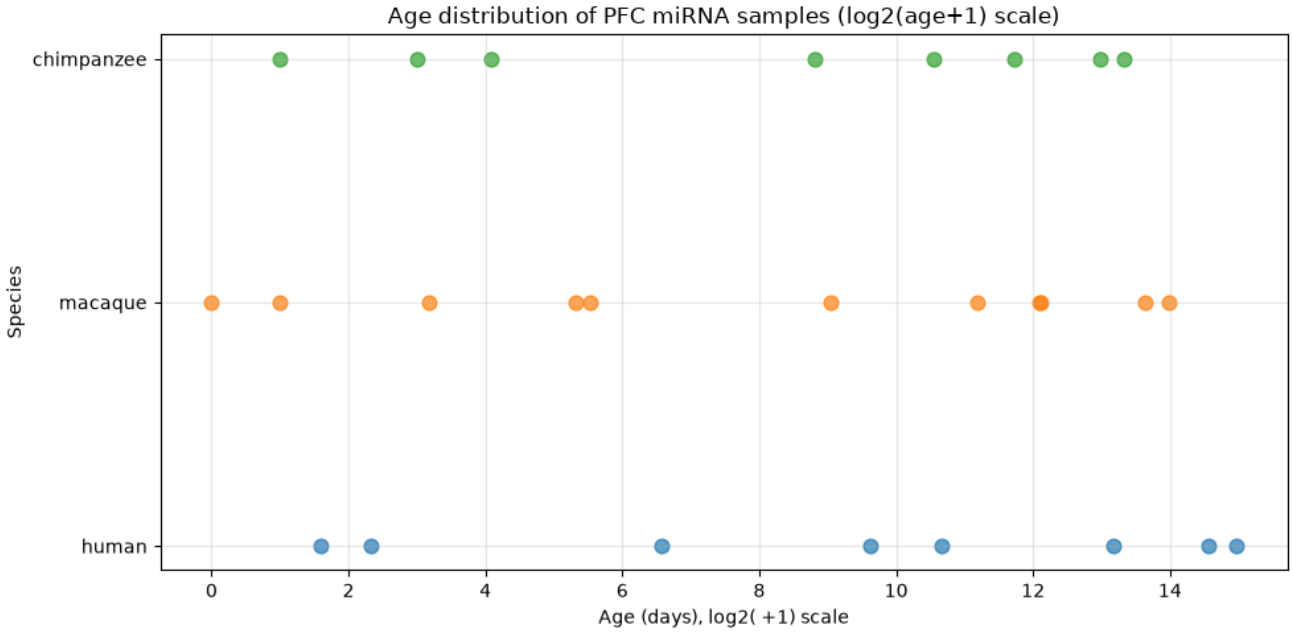
Age distributions

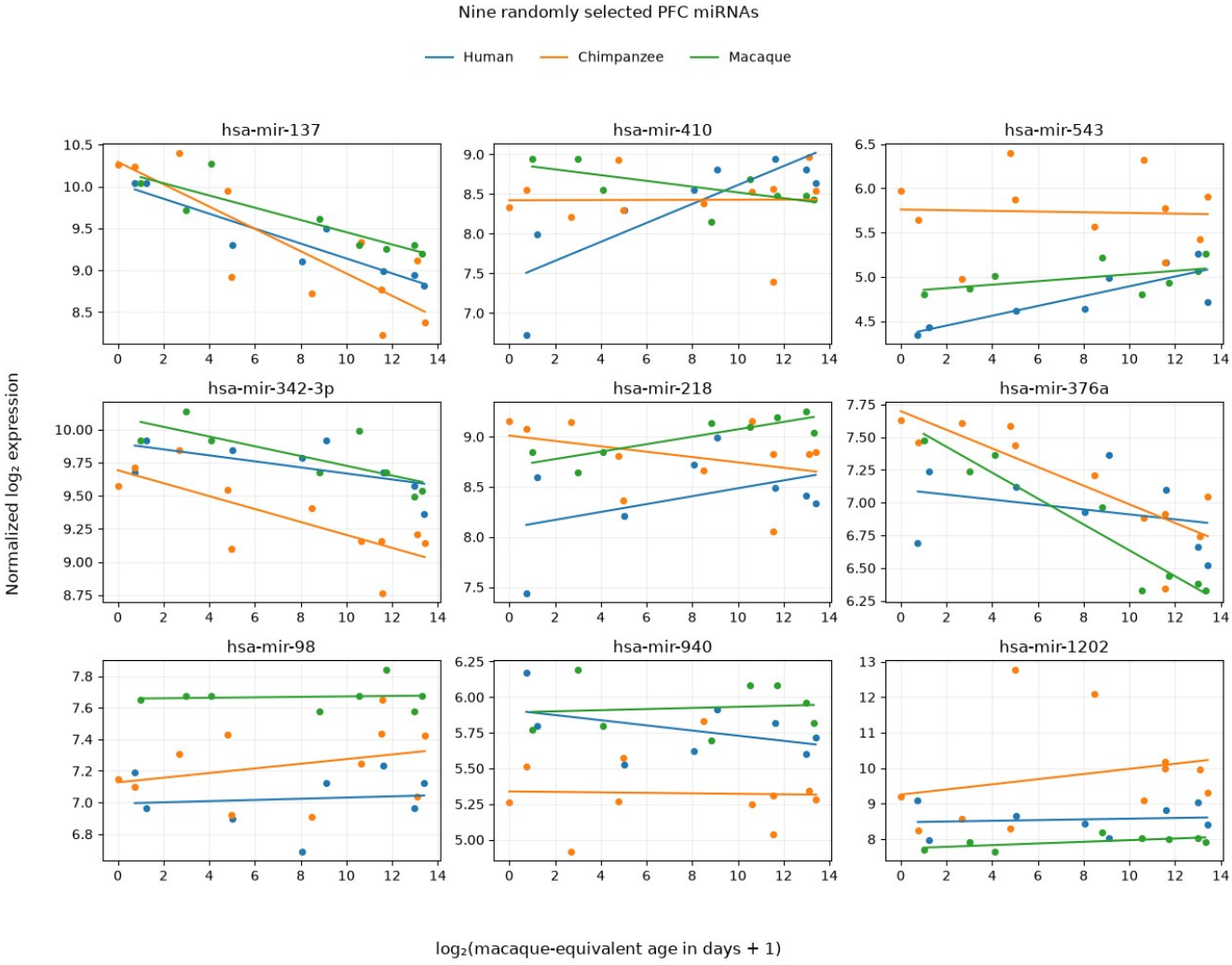
a) mRNA – PFC





b.) miRNA – PFC


Age distribution of PFC miRNA samples (log2(age+1) scale)
chimpanzee
macaque
human
Species
0
2
4
6
8
10
12
14
Age (days), log2(+1) scale


Nine randomly selected PFC miRNAs
Human
Chimpanzee
Macaque
hsa-mir-137
hsa-mir-410
hsa-mir-543
hsa-mir-342-3p
hsa-mir-218
hsa-mir-376a
hsa-mir-98
hsa-mir-940
hsa-mir-1202
Normalized log₂ expression
log₂(macaque-equivalent age in days + 1)

Week 3: miRNA–mRNA seed annotation and miRDB target matching

Goals: 1.) match historical PFC miRNAs to current seed-family annotations;

2.) construct a miRDB-based, high-confidence seed–target-gene network;

3.) document the limited coverage of the alternative target databases.

1.) Expression resources and analysis strategy

Data sources. The PFC miRNA expression data (GSE29356) contain historical mature-miRNA names. Human miRBase release 12 (miRBase12_human.csv) supplies their stable MIMAT accessions; the human annotation in TargetScan 8.0 (miR_Family_Info.txt) supplies updated names and seven-nucleotide Seed+m8 families. miRDB v6.0 supplies predicted miRNA–target interactions, and the GRCh38.p14 human RefSeq feature table supplies transcript-to-gene identifiers.

Primary network. TargetScan is used only for miRNA identifier and seed-family matching, not for predicted target edges. miRDB predictions are mapped to genes measured in the processed PFC mRNA dataset (GSE22521).

2.) Historical miRNA annotation and seed-family assignment

Notebook used: 4a_miRNA_seed_matching.ipynb

Identifier matching. The historical microarray labels, including * passenger-strand names, were matched to miRBase 12 and then to TargetScan 8.0 through stable MIMAT accessions. All 198 measured PFC miRNA names matched miRBase 12; 197 also received a current identifier and Seed+m8 annotation. These correspond to 157 distinct seed families.

Unmatched annotation. Only hsa-mir-886-3p (MIMAT0004906) lacked a TargetScan seed assignment. The resulting results/pfc_mirna_annotation.csv retains the original name, MIMAT accession, current miRNA name, descriptive family and seven-nucleotide seed. This file is the input to notebook 4b.

3.) Coverage limitations of TargetScan predictions and miRTarBase

TargetScan prediction coverage. In the earlier exploratory TargetScan default human target-prediction file, 49 of the 157 measured seed families (31.2%, approximately one third) had no predicted entries. This does not mean the seeds lacked TargetScan family annotations; it is a limitation of that particular prediction file.

miRTarBase coverage. The earlier, strict miRTarBase 10.0 subset (functional interactions with reporter-assay evidence) matched 177 measured miRNAs and only 3,786 distinct mature-miRNA–gene edges. This is too sparse to represent all plausible miRNA–mRNA interactions in this exploratory network analysis. Neither the TargetScan target predictions nor the restricted miRTarBase interaction set is used in the final miRDB network.

4.) miRDB high-confidence target prediction and matching

Notebook used: 4b_miRNA_targets_matching_miRDB.ipynb

Prediction source and cutoff. The compressed miRDB v6.0 input data/miRDB_v6.0_prediction_result.txt.gz contains predicted mature-miRNA–RefSeq transcript interactions. Filtering for human hsa- records with scores ≥80 yielded 812,240 predictions. Scores do not constitute experimental validation.

Transcript-to-gene mapping. Using GCF_000001405.40_GRCh38.p14_feature_table.txt.gz, 37,928 of 38,380 needed RefSeq accessions mapped to human genes (452 unmatched), covering 16,527 genes. The table mapped 807,022 high-confidence predictions to gene symbols and left 5,218 unmapped.

Measured miRNA matching. Eight miRNA-name version differences were reconciled before joining predictions to the measured-miRNA seed annotation. The join retained 90,081 prediction records for 195 measured miRNAs, covering 156 seed families and 10,318 mapped genes before PFC-gene restriction. Two seed-annotated miRNAs lacked a matching miRDB prediction: hsa-mir-137 and hsa-mir-320a.

Final PFC seed–gene network. We restricted target genes to the 11,667-gene PFC mRNA matrix and collapsed transcript- and mature-miRNA-level records into unique (Seed, Gene) pairs. Each edge retains the maximum miRDB score and the names and number of contributing measured mature miRNAs. The final graph contains 24,111 pairs, 7,039 target genes and 156 seed families, with no duplicate edges.

miRDB: selection, matching and network counts

Selection stage	Result / count
miRDB v6.0 predictions (all species)	6,831,595 prediction records
Human (hsa-) predictions	3,375,741 prediction records
Human predictions with miRDB score ≥80	812,240 predictions; 2,638 miRNAs; 38,380 RefSeq transcripts
RefSeq records assigned to human genes	37,928 RefSeq accessions; 16,527 genes (452 accessions unmatched)
High-confidence predictions mapped to genes	807,022 mapped; 5,218 unmapped
Measured PFC miRNAs with seed annotation	197 mature miRNAs; 157 distinct seeds
Predictions after measured-miRNA matching	90,081 records; 195 miRNAs; 156 seeds
Measured, seed-assigned miRNAs without matched predictions	2: hsa-mir-137 and hsa-mir-320a
Target genes after miRNA matching (before PFC restriction)	10,318 genes
Final network after PFC-gene restriction	**24,111 seed–gene edges; 7,039 genes; 156 seeds; 0 duplicate edges**

Interpretation. The vertices used for the later network analysis are measured seed families and their predicted target genes. Different mature miRNAs can share a seed. The matched_miRNAs and n_matched_miRNAs fields preserve their identities without duplicating seed–gene edges.

5.) Files saved during seed annotation and miRDB matching

Analysis files. The two notebooks export the following CSV files to results/. The first two document identifier matching; the final seed–gene network is used with the expression matrices and sample metadata in Week 5.

Saved analysis files (relative to the results/ directory)

Filename	Contents / purpose
pfc_mirna_annotation.csv	4a output: 198 measured historical miRNA names, MIMAT accessions, updated names and seven-nt Seed+m8 families.
mirdb_refseq_to_gene_symbol.csv	4b output: RefSeq-to-gene mapping for 37,928 accessions, covering 16,527 genes.
pfc_network_mirdb_high_confidence.csv	4b output: 24,111 unique seed–gene edges (156 seeds, 7,039 genes); columns Seed, Gene, miRDB_score, matched_miRNAs and n_matched_miRNAs.

Week 4: Early human–chimpanzee PFC differential expression

Goals: 1.) Find genes expressed differently between humans and chimpanzees during early development.

2.) Test whether autism-associated genes are overrepresented among the identified genes.

3.) Profile the functions of the identified genes using Gene Ontology (GO).

Progress recorded here: data selection, differential expression (Goal 1), SFARI autism enrichment (Goal 2), and Gene Ontology functional profiling (Goal 3).

1.) PFC mRNA gene-expression cohort

Notebook used: 1b_PFC_mRNA_human_specific.ipynb

Starting data and annotation. We used the processed GSE22521 prefrontal cortex mRNA profiles and GPL6244 gene annotation prepared in Week 2. The gene-level matrix contains 11,667 genes, each represented by one uniquely assigned microarray feature. The processed expression values were already log2-transformed and normalized; they were not transformed again.

Cohort selection. Notebook 1b selected 24 postnatal individuals with species-adapted age below **1,095 days**: 8 humans, 5 chimpanzees and 11 macaques. It **retained** the 2-day human GSM559193 and 19-day human GSM559186, which had been flagged by the earlier correlation screen, while excluding the technical repeat GSM559194 and prenatal macaques GSM559214 and GSM559225. This is an exploratory cohort choice.

Species	Selected 1b profiles	Age multiplier	Early test (Age_days < 100)
Human	8	1.000	5: 2, 4, 19, 34, 94 days
Chimpanzee	5	2.025	4: 0, 1, 8, 39 days
Rhesus macaque	11	2.970	Not included in this test
Total	24	—	9 early human/chimpanzee profiles

Life-history alignment. Species-adapted age = chronological age × species multiplier; chimpanzee multiplier = (30/44)/(33/98) and macaque multiplier = 1/(33/98). The analysis uses log2(species-adapted age + 1) as a continuous predictor, but applies the early-development cutoff to chronological age.

Input files from 1b: results/pfc_mrna_gene_expression_human_spec.csv (11,667 genes × 24 samples) and results/pfc_mrna_selected_sample_info_human_spec.csv (24 metadata records). The nine-sample early subset is formed within the subsequent pylimma notebook.

2.) Differential-expression model and selection rules

Notebook used: 6_PFC_early_dev_human_chimp_diff.ipynb

Chimpanzee is the reference species; a common age slope is fitted to both species.

For each gene g, the model is $Y_{gi} = \alpha_g + \beta_g [\log2(\text{adapted age}_i + 1) - \text{mean}(\log2(\text{adapted age} + 1))] + \delta_g I(\text{Human}_i) + \varepsilon_{gi}$. Here δ_g is the age-adjusted human-minus-chimpanzee log2 expression difference. Centering is over the nine included individuals.

Variance moderation and significance. The model was fitted gene by gene using pylimma.lm_fit, followed by empirical-Bayes moderation with trend=True and robust=True. The primary pylimma.treat test evaluates whether |δ_g| exceeds 0.30 log2 units; the Benjamini–Hochberg adjustment is applied across all 11,667 genes. The standardized effect is δ_g divided by the posterior residual standard deviation.

Stability. Each of the nine individuals was omitted in turn and the common-slope model refitted. The coefficient sign must agree with the complete-data sign in at least eight of nine refits. The same full-data centering is retained in these fits.

Parameter	Value / role
AGE_CUTOFF_DAYS	100; retain Human and Chimpanzee with Age_days < 100
MIN_ABS_LOG2_DIFFERENCE	0.30; minimum absolute adjusted difference and TREAT threshold
MAX_TREAT_FDR	0.15; strict upper limit on BH-adjusted TREAT p-value
MIN_ABS_STANDARDIZED_EFFECT	0.80; minimum absolute posterior-SD-standardized effect
MIN_LOO_DIRECTION_FRACTION	8 / 9; at least eight matching signs across nine refits
MAX_PROFILE_GENES	12; upper bound for displayed gene profiles

Gene selection. For each gene, the age-adjusted species coefficient was estimated by a common-slope linear regression. Empirical-Bayes moderation stabilizes gene-wise residual variance estimates using information across all tested genes. The TREAT test asks whether the absolute species difference exceeds 0.30 log2 units; Benjamini–Hochberg adjustment controls the false-discovery rate across 11,667 gene tests. The final candidates must also satisfy the effect-size and stability thresholds.

Minimum absolute posterior-SD-standardized effect (0.80). We divided the fitted human-minus-chimpanzee coefficient by the empirical-Bayes posterior residual standard deviation for that gene. A value of at least 0.80 in absolute magnitude means that the adjusted species difference is at least 0.8 times this residual standard deviation. It measures effect size, not a second p-value.

Leave-one-out direction (8/9). We refitted the regression nine times, each time excluding one individual. At least eight of the nine resulting species coefficients had to retain the same sign as the coefficient fitted to all nine individuals. This checks the direction of the effect; each refit was not required to remain statistically significant.

3.) Early human–chimpanzee results

Outcome	Result
Genes tested	11,667
Genes meeting LOO direction criterion	9,421
Genes passing all selection criteria	140
Selected genes higher in humans	62
Selected genes higher in chimpanzees	78

The 140 genes (about 1.2% of tested genes) passed all four requirements: TREAT FDR < 0.15, absolute adjusted difference ≥ 0.30, absolute standardized effect ≥ 0.80, and concordant direction in at least 8 of 9 leave-one-out fits. Among the selected genes, 62 have positive and 78 negative human-minus-chimpanzee coefficients.

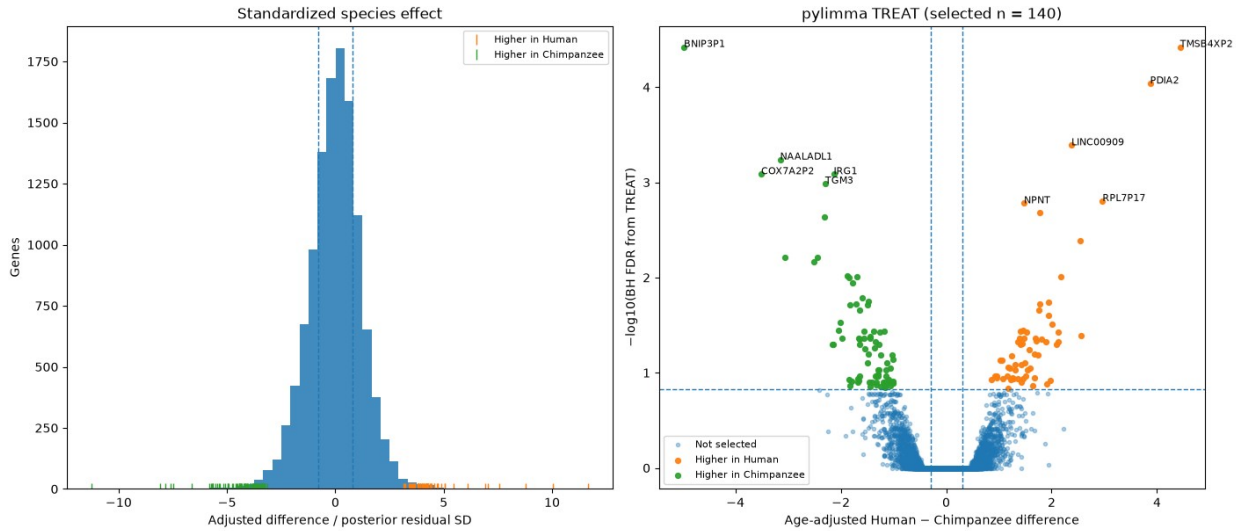


Figure 1. Distribution of standardized species differences and TREAT FDR versus the adjusted log2 difference (pylimma notebook output).

4.) Files produced by the pylimma analysis

File in results/	Purpose
early_human_chimp_pylimma_results_human_spec.csv	All 11,667 gene-level estimates, TREAT p-values, FDR and leave-one-out summaries
early_human_chimp_pylimma_candidates_human_spec.csv	140 genes passing all requested criteria
early_human_chimp_pylimma_sample_table_human_spec.csv	Nine early samples and centered age covariate
early_human_chimp_pylimma_distribution_human_spec.png	Standardized-effect histogram and TREAT FDR distribution

5.) SFARI autism-gene enrichment among early human–chimpanzee differences

Notebook used: 7_PFC_early_dev_autism_genes_enrichment_with_Fisher.ipynb

Gene sets and background. We matched SFARI gene symbols to the 11,667 genes tested by pylimma, using uppercase exact-symbol matching. Of 245 SFARI Category 1 genes, 217 were represented in the tested set; of 957 Category 1 + 2 genes, 732 were represented. We compared the 140 selected genes with the remaining 11,527 tested genes.

Testing procedure. We performed two one-sided Fisher exact tests for overrepresentation (alternative = greater): Category 1 high-confidence autism genes and Categories 1 + 2 combined. The tables show the observed gene counts. The reported p-values are not adjusted for testing both sets.

SFARI Category 1 (high-confidence autism genes)

Gene group	Category 1 gene	Other tested gene
Selected (140)	0	140
Not selected (11,527)	217	11,310

Fisher exact test (one-sided enrichment): p = 1.0000; odds ratio = 0.000.

SFARI Categories 1 + 2 (high-confidence and strong-candidate genes)

Gene group	Category 1 + 2 gene	Other tested gene
Selected (140)	4	136
Not selected (11,527)	728	10,799

Fisher exact test (one-sided enrichment): p = 0.978922; odds ratio = 0.4363.

Interpretation. No Category 1 gene and only four Category 1 + 2 genes were found among the 140 selected genes. These overlaps are below the expectations under no association (approximately 2.6 and 8.8 genes, respectively). Both odds ratios are below 1, pointing descriptively toward underrepresentation, not enrichment. Neither one-sided test provides evidence for overrepresentation; these tests alone do not establish statistically significant underrepresentation. Exact symbol matching may miss unconverted gene aliases.

6.) Gene Ontology analysis of early human–chimpanzee expression

Notebook used: 8_PFC_GeneOntology_analysis.ipynb

Method. Using GOfuncR in R, we ranked all 11,667 genes by the signed pylimma moderated t-statistic and tested GO Biological Process terms with the Wilcoxon method. Higher ranks indicate higher age-adjusted expression in humans; lower ranks indicate higher expression in chimpanzees. Permutation-based family-wise error rates (FWER) correct for testing many related GO terms. The completed run used 10,000 random sets and FWER <= 0.05.

Annotation coverage. GOfuncR accepted 10,386 of the 11,667 tested genes and evaluated 13,796 Biological Process terms. It identified 76 human-higher and 106 chimpanzee-higher significant terms before GO hierarchy refinement.

Analysis	Genes with GO annotation	Terms at FWER <= 0.05
All-gene ranking: human-higher	10,386	76
All-gene ranking: chimpanzee-higher	10,386	106
Selected-gene set: hypergeometric	116 selected genes	0

Selected-gene analysis. A separate hypergeometric test compared the 140 selected genes with the tested-gene background. Of the selected genes, 116 had usable GO annotations. No GO term was significantly overrepresented at FWER <= 0.05. Thus, the ranked genome-wide analysis showed functional associations not detected by the thresholded 140-gene test.

GO hierarchy refinement. GOfuncR refine uses an elim-like procedure: contributions from significant child terms are removed before retesting broader parent terms. This reduces redundancy and highlights more specific biological processes.

Expression direction	Original significant terms	Retained after refinement	Representative themes
Human-higher	76	10	Transcriptional regulation; RNA splicing; DNA-related processes
Chimpanzee-higher	106	30	Synaptic signaling and plasticity; ion transport; mitochondrial electron transport

Refined categories (10,000-random-set analysis). The tables below give all 10 human-higher and 30 chimpanzee-higher GO Biological Process categories retained after the hierarchy refinement (signif = TRUE in the exported GOfuncR files). FWER values are from the original ranked Wilcoxon test, before refinement.

Human-higher refined categories (10 terms)

GO ID	Refined Biological Process category	Original FWER
GO:0019219	regulation of nucleobase-containing compound metabolic process	<0.0001
GO:2000112	regulation of cellular macromolecule biosynthetic process	<0.0001
GO:0006357	regulation of transcription by RNA polymerase II	<0.0001
GO:0000398	mRNA splicing, via spliceosome	0.0004
GO:0019827	stem cell population maintenance	0.0025
GO:0006974	cellular response to DNA damage stimulus	0.0054
GO:0006259	DNA metabolic process	0.0099
GO:0098813	nuclear chromosome segregation	0.0107
GO:0045892	negative regulation of transcription, DNA-templated	0.0169
GO:0006323	DNA packaging	0.0225

Chimpanzee-higher refined categories (30 terms)

GO ID	Refined Biological Process category	Original FWER
GO:0007416	synapse assembly	0.0488
GO:0042775	mitochondrial ATP synthesis coupled electron transport	0.0391
GO:0031644	regulation of nervous system process	0.0384
GO:0019646	aerobic electron transport chain	0.0366
GO:0070588	calcium ion transmembrane transport	0.0278
GO:0007215	glutamate receptor signaling pathway	0.0273
GO:0030073	insulin secretion	0.0273
GO:0050905	neuromuscular process	0.0170
GO:0048667	cell morphogenesis involved in neuron differentiation	0.0108
GO:0016082	synaptic vesicle priming	0.0107
GO:0060996	dendritic spine development	0.0083
GO:0060291	long-term synaptic potentiation	0.0081
GO:0006874	cellular calcium ion homeostasis	0.0076
GO:0035249	synaptic transmission, glutamatergic	0.0053
GO:0099505	regulation of presynaptic membrane potential	0.0047
GO:1902600	proton transmembrane transport	0.0046
GO:0007186	G protein-coupled receptor signaling pathway	0.0044
GO:0001508	action potential	0.0042
GO:0046928	regulation of neurotransmitter secretion	0.0034
GO:0017157	regulation of exocytosis	0.0020
GO:0007613	memory	0.0010
GO:0007612	learning	0.0010
GO:0048812	neuron projection morphogenesis	0.0009
GO:1904062	regulation of cation transmembrane transport	<0.0001
GO:0071805	potassium ion transmembrane transport	<0.0001
GO:0060078	regulation of postsynaptic membrane potential	<0.0001
GO:0048167	regulation of synaptic plasticity	<0.0001
GO:0099504	synaptic vesicle cycle	<0.0001
GO:0050808	synapse organization	<0.0001
GO:0007268	chemical synaptic transmission	<0.0001

FWER note. Values displayed as <0.0001 indicate zero exceedances at the resolution of the 10,000 random-set procedure; they are not exact zero probabilities. Original FWER is shown for context; the refined categories are those flagged as significant by GOfuncR after hierarchical refinement.

Interpretation. Human-higher expression was associated mainly with transcriptional and RNA/DNA-related functions, whereas chimpanzee-higher expression emphasized neural signaling, ion handling and mitochondrial processes. These are associations with expression-rank shifts across many genes, not evidence that a process is absent in either species or causally altered.

Final notebook. 8_PFC_GeneOntology_analysis.ipynb reads results/early_human_chimp_pylimma_results_human_spec.csv and uses N_RANDSETS = 10000, RANDOM_SEED = 42 and FWER_CUTOFF = 0.05. The executed analysis reported 76 human-higher and 106 chimpanzee-higher GO terms at FWER <= 0.05, with 10 and 30 retained after hierarchy refinement.

Final GO output files. The finalized notebook exports the ranked and selected-set GO tables to results/gofuncr_ranked_all_terms_human_spec.csv, results/gofuncr_ranked_significant_human_spec.csv, results/gofuncr_selected_all_terms_human_spec.csv and results/gofuncr_selected_significant_human_spec.csv; retained refined terms are saved as results/gofuncr_refined_human_significant.csv and results/gofuncr_refined_chimp_significant.csv. The ranked result object is results/gofuncr_ranked_complete.rds.

Week 5: miRNA–mRNA network analysis with miRDB

Goals: 1.) Construct sample-specific miRNA–mRNA network activity scores from matched PFC expression profiles.

2.) Compare developmental trends in network-level features between humans and chimpanzees.

3.) Export gene-level network scores for subsequent analysis.

1.) Input data, sample matching and network

Notebook used: 5_miRNA_mRNA_network_analysis_miRDB.ipynb

Source data. The analysis uses the processed PFC miRNA and mRNA expression matrices (GSE29356 and GSE22521), the sample metadata, historical miRNA-to-seed annotation and the high-confidence miRDB seed–gene interaction file. It aligns miRNA and mRNA profiles by species and original chronological age. A one-day offset in processed miRNA sample names is corrected using Original_age_days.

Matched species	Profiles (n)	Chronological ages represented (days)
Human	7	4 – 32,120
Chimpanzee	10	0 – 16,131
Rhesus macaque	7	1 – 10,220
Total	24	Across infancy through adulthood

Matching note. Three miRNA profiles had no match in the retained mRNA set (human, 2 days; chimpanzee, 45 days; macaque, 1,487 days). Table S1 supports substantial overlap between subjects assayed for PFC miRNA and mRNA, but notebook 5 joins profiles by species and age, not verified donor identifiers. The 24 matched profiles include 17 human and chimpanzee profiles used for inference.

Network construction. Each mature-miRNA and gene expression trajectory was standardized across all 24 matched profiles (row-wise z-scores). The 197 seed-annotated mature miRNAs were averaged within 157 seed families after standardization. Restriction to measured seeds and genes yielded the final network: 156 seed families, 7,039 target genes and 24,111 distinct seed–gene edges.

Network consistency. The analysis uses all 24 profiles, including seven macaques, to calculate feature-wise standardization parameters and weighted network features. Macaques are shown as descriptive comparators, but the interaction and permutation tests use only 7 human and 10 chimpanzee profiles.

2.) Construct the edge scores and six network features

For every profile and predicted miRNA-seed → mRNA-target edge, an activity score was computed from standardized expression as sigmoid(z_seed − z_gene) = 1 / [1 + exp(−(z_seed − z_gene))]. High scores indicate relatively high seed expression and low target expression. These scores are expression-derived proxies, not direct measurements of miRNA-mediated repression. Scores are summarized across the same predicted network for all 24 profiles.

Network feature	Definition (per profile)
Mean edge score	Average logistic activity score across all seed–target edges.
Active-edge fraction	Proportion of edge scores ≥ 0.75.
Edge-score dispersion	Standard deviation of scores across all network edges.
Family out-strength CV	SD/mean of mean edge scores, computed across seed families.
Target in-strength CV	SD/mean of mean incoming edge scores, computed across target genes.
Family out-strength entropy	Normalized Shannon entropy of seed-family mean edge scores.

Age alignment. Developmental time was converted to macaque-equivalent days using human × (33/98), chimpanzee × (30/44) and macaque × 1. The analysis used log2(macaque-equivalent age + 1). This alignment is approximate and intended to compare life-history stages rather than equal chronological ages.

3.) Human–chimpanzee developmental interaction tests

Statistical model. Each network feature was regressed on species, log2-aligned age and the species × age interaction: feature = intercept + species + aligned age + species × aligned age + error. Chimpanzee is the reference species; the interaction coefficient is the human-minus-chimpanzee slope difference. Freedman–Lane tests permuted the residuals of the reduced model without interaction 5,000 times (seed 29356), refitted the full model and computed two-sided permutation p-values. Benjamini–Hochberg FDR correction was applied across the six features.

Feature	Human − chimp slope	Perm. p	BH FDR	R^2
Target in-strength CV	−0.010028	0.002200	0.013197	0.526
Edge-score dispersion	−0.005435	0.015197	0.045591	0.444
Family out-strength CV	−0.010393	0.142372	0.242951	0.240
Family out-strength entropy	+0.000878	0.161968	0.242951	0.212
Active-edge fraction	−0.006396	0.250550	0.300660	0.182
Mean edge score	+0.001407	0.655469	0.655469	0.058

Results. Two of six features passed BH FDR < 0.05: target in-strength CV (interaction t = −3.713, permutation p = 0.002200, FDR = 0.013197) and edge-score dispersion (t = −2.742, p = 0.015197, FDR = 0.045591). The respective human and chimpanzee slopes were −0.006979 and +0.003048 (target in-strength CV), and −0.004600 and +0.000835 (edge-score dispersion). The other four interaction tests were not significant.

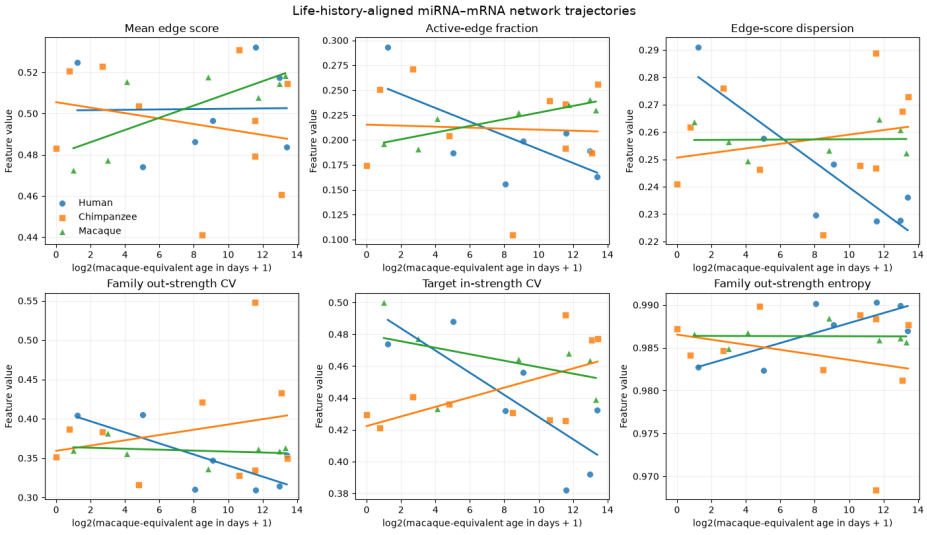


Figure 2. Six sample-level miRNA–mRNA network features across log2 life-history-aligned age, including descriptive linear fits for human, chimpanzee and macaque profiles (executed notebook output).

4.) Interpretation and exported file

Interpretation and exports. The two significant trends concern expression-derived network scores, not direct repression. Notebook 5 saves per-profile features (results/pfc_network_sample_features_mirdb.csv), permutation tests (results/pfc_network_permutation_tests_mirdb.csv), the six-panel figure (results/pfc_network_six_features_trajectories.png), and gene-level scores (results/gene_level_network_scores_mirdb.csv; 168,936 rows = 7,039 genes × 24 profiles).